CivicVoice AI

Multilingual Voice-Based Civic Complaint Assistant

Problem

Reporting everyday civic issues can be difficult when complaint processes require citizens to type detailed descriptions or navigate complex digital forms. This can create barriers for people who prefer speaking or are less comfortable with text-based interfaces.

Our Solution

CivicVoice AI is a prototype that lets citizens report civic issues through voice. It converts speech into text, assigns a civic-issue category and priority using a keyword-based classifier, records the complaint for administrative review, and can provide an audio confirmation.

How It Works

1. **Voice input:** A citizen records or uploads a complaint and selects English or Kannada.
2. **Speech recognition:** Gnani AI Prisma STT transcribes supported speech into text.
3. **Classification:** The prototype uses keyword rules to assign a category and priority.
4. **Registration:** A unique complaint ID, description, category, priority, timestamp, status and location are stored when provided.
5. **Audio confirmation:** Gnani AI Timbre TTS generates a spoken confirmation (currently tested with the configured Hindi voice).
6. **Administration:** The Gradio dashboard supports complaint search and filtering, status updates, audit history, charts, configurable SLA monitoring, possible-duplicate detection and CSV exports.

Technology

Python, Gradio, Pandas, Plotly, Google Drive, Gnani AI Prisma Speech-to-Text and Gnani AI Timbre Text-to-Speech.

Validation

The administrator backend passed 17/17 diagnostic checks covering database access and persistence, complaint creation and status updates, audit logging, SLA calculations, similarity-based duplicate detection, chart generation, search, filters and CSV exports.

Impact and Next Steps

CivicVoice AI aims to make civic issue reporting more accessible and help administrators organize incoming reports. Future work includes integrating an AI reasoning model, expanding verified multilingual speech output, connecting to a secure production database, and integrating with authorized civic-service systems.

Prototype scope: This build records complaints for review; it does not submit them to government authorities. Keyword classification is not an AI reasoning model. The dashboard's 24-hour and 48-hour SLA thresholds are configurable prototype settings, not official service standards. The citizen app and administrator dashboard should only be described as sharing live records if that end-to-end connection has been tested.

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